

Succinct Non-interactive Arguments for Distributions

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Abstract. We study a new notion of succinct non-interactive arguments (SNARG) for distributions, where the prover is given the full density function of the distribution, while the verifier has only sample access to the distribution. This notion can be viewed as an analogue of distributional property testing in the context of (non-interactive) computationally sound proofs. We obtain the following results:

- **SNARGs for NP-Distribution Properties.** Assuming indistinguishability obfuscation and learning with errors (LWE), we obtain the first non-adaptively sound SNARG for any distribution properties in NP. For distributions supported on $[n]$, our proof size and verifier sample complexity are both $O(n^{2/3})$. Moreover, our SNARG is zero-knowledge—the proof can be simulated in polynomial time using $O(n^{2/3})$ samples from the distribution.
- **SNARKs for P-Distribution Properties.** Assuming simple standard assumptions such as LWE, or subexponential DDH, or bilinear maps, we construct the first non-adaptively sound SNARK for any distribution properties over $\Omega(1/n)$-grained distributions in P. Our construction achieves proof size $O(n^c)$ and sample complexity $O(n^c)$ for some constant $c \in (0, 1)$. Along the way, we also construct the first succinct non-interactive batch arguments for distribution properties in NP under the same assumptions.
- **Lower Bound.** Assuming LWE, we prove a lower bound of non-adaptively sound SNARGs for P/poly properties over distributions supported on $[n]$, showing that the proof size is $\tilde{\Omega}(n^{3-4\delta})$ when the verifier's sample complexity is $O(n^\delta)$ where $\delta \in (1/2, 2/3)$.

To obtain our results, and of independent interest, we introduce a notion of sketching algorithms for distributional identity testing. We prove both upper and lower bounds for such sketching for distributions. Our techniques employ distribution property testing and communication complexity lower bounds.

1 Introduction

Succinct non-interactive arguments (SNARGs) [34] allow a prover to convince a verifier of the truth of a statement via a short, one-round message. The stan-

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dard setting of SNARGs is the common reference string (CRS) model. They have a wide range of applications, ranging from delegation [15] and verification of distributed computation [8] to zero-knowledge [32] and verifiable private information retrieval [2]. Recent years have seen significant progress in constructing SNARGs under standard cryptographic assumptions [9,11,24,27–29,31,39], as well as extensions of the model that allow verification in sublinear time via SNARGs of proximity [6,30] or holographic SNARGs [33].

Despite this progress, all existing SNARG models typically assume that the statement to be verified is given as an explicit bit string. However, in many practical scenarios, the statement itself expresses a claim about a distribution D, while the verifier may not have access to any bit representation of D. Instead, the verifier may only be able to sample from D. For example, scientific experiments often involve collecting samples from the distribution of a natural phenomenon, such as the distribution of real-world images, physical measurements, or other observable data in nature. In such scenarios, verification must proceed using only sample access to the underlying distribution, rather than explicit knowledge of its density function. Another motivating example is the verification of large language models (LLMs). Suppose a prover, who has access to the model parameters, wishes to convince a verifier that the model satisfies certain properties, such as compliance with a safety standard. The verifier, however, might only be able to access samples from the model's output distribution, and this minimal assumption is the one we adopt.[1]

These examples motivate a new variant of SNARGs, which we call *SNARGs for distributions*. In this setting, the prover is given the full density function of a distribution D over a finite universe $[n] = 1, 2, \ldots, n$, whereas the verifier only has oracle access that allows drawing samples from D. Since a small perturbation in the density function is unlikely to be detected by such a verifier, we relax the standard SNARG soundness definition to tolerate an ε-error in statistical distance. That is, the soundness of SNARGs for distributions for a language $\mathcal{L}$ only requires that, for any statement that is ε-far from all distributions in $\mathcal{L}$, any polynomial-time cheating prover can only convince the verifier to accept the proof with negligible probability. This notion of SNARGs can be regarded as the natural analogue of distribution property testing in the context of non-interactive cryptographic proofs.

We focus on SNARGs for distributions that achieve both sublinear proof size and sublinear sample complexity for the verifier. This leads to the following fundamental question:

Can we construct SNARGs for distributions—under what assumptions, and with what proof size and verifier sample complexity?

Prior Work. There has been an exciting recent line of work that studies interactive proof systems for distributions. The most relevant to our work is the

[1] Alternative approaches exist, such as preprocessing the model parameters to produce a hash value so that the verifier only needs the hash to verify the proof. But this introduces an additional assumption that the hash value can be computed honestly.

recent elegant work of [17], which constructs a 4-round interactive argument for general distributions with total communication complexity and sample complexity $\tilde{O}(\sqrt{n}/\varepsilon^2)$, based on collision-resistant hash functions. In contrast, our work focuses on the non-interactive setting, as in SNARGs, which we view as conceptually the simplest. Other prior works on proof systems for distributions have primarily considered the statistical soundness setting, which requires the soundness to hold even against computationally unbounded cheating provers.

In the setting of non-interactive proofs with statistical soundness, the work [7] initiated the study of non-interactive proof (MA) of proximity for distributional properties. They showed that every property of the distributions over $[n]$ has an MA tester with proof size $\tilde{O}(n)$ and sample complexity $O(\sqrt{n})$. They also proved a lower bound: for any MA tester with proof size p and sample complexity q, it must hold that $p \cdot q = \Omega(s_{\Pi})$ where s_{Π} is the sample complexity of the property Π without proofs. However, it remains open whether any barrier exists in the computationally sound setting. In the case of uniformity testing, they show a lower bound of $\Omega(\sqrt{n})$ sample complexity based on the lower bound of uniformity testing [36], no matter how long the proof is. This lower bound also extends to the computationally sound setting.

For statistically sound interactive proofs for distributional properties, [19] improves upon [18], giving a doubly-efficient 2-round interactive proof for label-invariant distribution properties, with communication complexity and sample complexity $\tilde{O}(\sqrt{n})$. The recent work [16] constructed a 2-round public-coin protocol with communication and sample complexity $\tilde{O}(n^{2/3})$ for label-invariant properties. More recently, [20] gave the first doubly-efficient interactive proofs for any property decidable by bounded-depth circuits (or bounded-space machines), with verifier cost $\tilde{O}(n^{1-\alpha})$ for some constant $\alpha > 0$.

In summary, all previous work focused on the interactive setting or the statistical soundness setting. Whether we can construct SNARGs for distributions remains open. We defer more related work to the full version.

2 Our Result

Before presenting our results, we first formally define distribution properties in P and NP. Our definitions are the natural extensions of *approximately decidable properties* in [18–20] to P and NP.

Distribution Properties in P *and* NP. A distribution property $\mathcal{L}$ is a set of distributions (similar to languages), where each distribution is defined over a finite universe $[n]$ for some integer n. To allow Turing machines to take distributions as input, we need to specify how distributions can be represented as binary strings. Following [14,20], we say a distribution D over $[n]$ is μ-*grained* for a real $\mu \in (0, 1]$, if its density mass $D[i]$ is a multiple of μ for every $i \in [n]$. Throughout this work, we will consider $1/\mathsf{poly}(n)$-grained distributions, which can be represented as $\tilde{O}(n)$ bits.

Let $\rho = \rho(n) \geq 1/\mathsf{poly}(n)$ be a *precision parameter*. We say that a distribution property $\mathcal{L}$ is ρ-*approximately decidable in* NP if there exists a nondeterministic polynomial-time Turing machine M with the following behavior. The machine

M takes as input proximity parameters $\Delta \in (0,1)$, together with the density function of a $\mathsf{poly}(\rho(n))$-grained distribution $D = \{(i, D[i])\}_{i\in[n]}$, and nondeterministically decides whether to accept or reject. Correctness for M requires that for any integer n, any Δ satisfying $\Delta \geq \rho(n)$, and any $\mathsf{poly}(\rho(n))$-grained distribution D, the machine M accepts if $D \in \mathcal{L}$, and rejects if D is Δ-far from $\mathcal{L}$ in statistical distance.

The notion of approximately decidable distribution properties in P is defined analogously.

SNARGs for Distributions (SNARG-Dist). We introduce the notion of SNARGs for distribution properties (SNARG-Dist). A SNARG-Dist for a distribution property $\mathcal{L}$ (in NP or P) with precision $\rho = \rho(n) \geq 1/\mathsf{poly}(n)$ consists of a prover $\mathcal{P}$, a verifier $\mathcal{V}$, and a trusted setup algorithm $\mathcal{G}$, defined as follows.

- **Setup** $\mathcal{G}$: The trusted setup algorithm generates a common reference string CRS.
- **Prover** $\mathcal{P}$: Given the CRS, proximity parameter $\varepsilon \in (0,1)$, and a full description of a $\mathsf{poly}(\rho(n))$-grained distribution D over a finite domain $[n]$, the polynomial-time prover generates a proof π. In the case of distribution properties in NP, the honest prover is additionally given a nondeterministic witness w.
- **Verifier** $\mathcal{V}^D$: The verifier takes the proof π, the CRS, and proximity parameter $\varepsilon \in (0,1)$ as input, and is only given sampling access to a distribution D. During verification, it may draw samples from D, and finally decide whether to accept or reject the proof.

For any constants $\varepsilon \in (0,1)$ that satisfy $\varepsilon \geq \rho(n)$ for all sufficiently large n, we require

- **Completeness:** For any $\mathsf{poly}(\rho)$-grained distribution $D \in \mathcal{L}$, the honestly generated proof is accepted by the verifier, except with negligible probability.
- **(Non-adaptive) Soundness:** For any distribution D that is ε-far from $\mathcal{L}$ in statistical distance, no polynomial-time cheating prover can produce a proof that causes the verifier $\mathcal{V}^D$ to accept, except with negligible probability.

We defer the full formal definition of SNARG-Dist to the full version.

Zero-Knowledge. We introduce a new notion of zero-knowledge for SNARG-Dist. We say a SNARG-Dist is zero-knowledge if the honestly generated proof by the prover algorithm can be simulated by a polynomial-time simulator that is only given a *sublinear* number of samples from D. This notion of zero-knowledge is parameterized by the simulator's sampling complexity.

We defer its formal definition to the full version.

Theorem 1 (zk-SNARGs for Distributions, Informal Version). *Assuming the existence of indistinguishability obfuscation (iO) and polynomial hardness of Learning with Errors (LWE), there exists a non-adaptively sound* SNARG-Dist *for any distribution property* $\mathcal{L}$ *in* NP *with precision parameter* $\rho = \rho(n) \geq 1/\mathsf{poly}(n)$. *Our* SNARG-Dist *achieves the following parameters:*

- *Proof size* $\mathsf{poly}(\lambda) \cdot \tilde{O}(n^{2/3})$.
- *Verifier's sample complexity* $\mathsf{poly}(\lambda) \cdot \tilde{O}(n^{2/3})$.

Moreover, the SNARG-Dist *proof can be simulated by a zero-knowledge simulator that is given* $\tilde{O}(n^{2/3})$ *samples from the distribution in polynomial time.*

Our Theorem 1 requires iO, whose construction currently requires a combination of multiple cryptographic assumptions [25,26,37]. We then turn to simple, standard cryptographic assumptions such as LWE, and focus on constructing SNARG-Dist for P-distribution properties in this setting.

Theorem 2 (SNARKs for distribution properties in P, Informal Version). *Assuming somewhere extractable batch arguments, somewhere extractable hash functions, and correlation intractable hash functions, there exists a non-adaptively sound SNARK (*SNARK-Dist*) for any* $\Omega(1/n)$*-grained distribution property* $\mathcal{L}$ *in* P *with precision parameter* $\rho = \rho(n) \geq 1/\mathsf{poly}(n)$.

There exists a constant $c \in (0,1)$ *such that our construction achieves the following parameters.*

- *Proof size:* $\mathsf{poly}(\lambda) \cdot \tilde{O}(n^c)$.
- *Verifier's sample complexity:* $\mathsf{poly}(\lambda) \cdot \tilde{O}(n^c)$.

In particular, our construction can be obtained by assuming polynomial-hardness of LWE, or sub-exponential DDH, or DLIN over bilinear maps.

Knowledge Soundness for Distributions. Our SNARG-Dist in Theorem 2 additionally achieves *knowledge soundness for distributions*, and we therefore call it a SNARK-Dist. Intuitively, this property requires that any prover that makes the verifier accept must "know" evidence that the verifier's D is close to $\mathcal{L}$. More specifically, we require that there exists an efficient uniform extractor such that, for any constant $\varepsilon \in (0,1)$, given the verifier's distribution D, and any accepting proof, the extractor can efficiently recover a distribution D^* that is ε-close to D and $M_{\mathcal{L}}(\varepsilon, D^*)$ accepts. Here $M_{\mathcal{L}}$ is the approximate decider for $\mathcal{L}$.

Two remarks are in order. First, note that this notion of knowledge soundness is non-trivial even for P, since it could be that $M_{\mathcal{L}}(\varepsilon, \cdot)$ rejects verifier's D, while D is ε-close to $\mathcal{L}$. In such a scenario, a cheating proof may still make the verifier accept, and knowledge soundness guarantees that we can extract an "ε-approximate version" of D: namely, a distribution D^* such that $M_{\mathcal{L}}(\varepsilon, D^*) = 1$ and $\mathsf{SD}(D, D^*) \leq \varepsilon$. Hence, this D^* "certifies" that D is ε-close to $\mathcal{L}$. Second, we note that we allow the extractor to have full access to the verifier's distribution D, whereas the verifier only has sample access to D.

Batch Arguments for Distributions (BARG-Dist). As an intermediate step toward constructing SNARK-Dist for P (Theorem 2), we introduce a SNARG for batches of distribution properties, which we call *batch arguments for distributions* (BARG-Dist). In this setting, the prover is given a batch of distributions $D_1, \ldots, D_k$ and aims to convince the verifier that D_i is close to $\mathcal{L}$ for all $i \in [k]$,

where $\mathcal{L}$ is a distribution property in NP. The verifier, however, can only obtain samples from each of the distributions $D_1, \ldots, D_k$ to verify the proof. The goal is to achieve a proof size that is asymptotically smaller than $k \cdot |w_i|$ where w_i is the NP-witness length of $\mathcal{L}$. This notion can be viewed as an analogue of batch proof systems studied in SNARG constructions [3,10,11] and interactive proofs [38]. Given the broad applications of BARGs [2,11,12,29,31,35], we believe this new notion is of independent interest.

Theorem 3 (BARG-Dist from Simple Standard Assumptions, Informal Version). *Assuming correlation intractable hash functions and somewhere extractable batch arguments, there exists a batch argument for any distribution properties ρ-approximately decidable in* NP *(*BARG-Dist *for* NP*), where $\rho = \rho(n) \geq 1/\mathsf{poly}(n)$.*

For $k = k(\lambda)$ distributions $D_1, \ldots, D_k$ with NP *witness length $|w_i| = m$, and any $\varepsilon = \varepsilon(n) \geq \rho(n)$, our construction achieves the following parameters:*

- *Proof size: $m + k \cdot (n^{2/3} \cdot \varepsilon^{-4/3} + \mathsf{poly}(\lambda))$*
- *Verifier's sample complexity: $n^{2/3} \cdot \varepsilon^{-4/3} \cdot \mathrm{polylog}(n)$*

In particular, our BARG-Dist *for* NP *can be instantiated from LWE, or subexponential DDH, or DLIN over bilinear maps.*

Remark 1 (Semi-adaptive Somewhere Argument of Knowledge). Our BARG-Dist for NP (Theorem 3) also satisfies a *semi-adaptive somewhere argument of knowledge*, analogous to classical batch arguments for NP [3,10,11]. Specifically, there exists an efficient extractor such that, when the CRS is generated with an index $i^* \in [k]$ embedded, then for any accepting BARG-Dist proof and any adaptively chosen $D_1, \ldots, D_k$ from the adversary, any $\varepsilon \geq \rho(n)$, the extractor—given the CRS trapdoor—can output a $1/\mathsf{poly}(n)$-grained distribution D_{i^*} and a witness w such that $M_{\mathcal{L}}(\epsilon, D_{i^*}, w)$ accepts and D_{i^*} is ε-close to D_{i^*}. Here, $M_{\mathcal{L}}$ is the decider for $\mathcal{L}$.

Finally, we prove a lower bound of SNARGs for distributions:

Theorem 4 (Lower bound of SNARGs for Distributions, Informal Version). *Assuming polynomial hardness of LWE, for any constant $\varepsilon \in (0, 1)$ and any $\delta \in (1/2, 2/3)$, any SNARGs for distribution properties in* P/poly *with verifier sample complexity $O(n^{\delta})$ must have proof length at least $\tilde{\Omega}(n^{3-4\delta})$.*

When $\delta = 2/3$—corresponding to the verifier's sample complexity $O(n^{2/3})$ in our construction—the lower bound yields only $\Omega(n^{1/3})$ on the proof length. This reveals a gap between our upper and lower bounds. Whether there is a construction of SNARGs for distributions with a smaller proof size and sample complexity, or whether the lower bound can be improved, remains an open question for future work.

2.1 More Discussions

SNARGs of Proximity. A line of previous work considered non-interactive proof systems for property testing, where the statement is not a distribution, but instead a binary string. The prover has full access to the statement, while the verifier can only query some locations of the statement. The goal is still to have the prover convince the verifier that the statement is close to an NP language. The previous work [30] showed the first 2-round argument of proximity for P with sublinear communication and verifier query complexity, which can also be viewed as a designated-verifier SNARG of proximity. The recent work [6] constructed a non-adaptively sound SNARG of proximity (SNAP) for NP from iO and LWE with $\mathsf{poly}(\lambda)$ proof size and verifier sample complexity, and adaptively sound SNAP for P from simple standard assumptions such as LWE, with $\widetilde{O}_\lambda(\sqrt{n})$ proof size and sample complexity. In the adaptively sound setting, they also proved an information-theoretical lower bound of $\Omega(\sqrt{n})$ proof size when verifier's sample complexity is $O(\sqrt{n})$.

We note that all of our positive and negative results only consider the non-adaptive soundness setting. Hence, the lower bound in the adaptively sound setting of [6] does not apply here. We leave the feasibility of adaptively sound SNARGs for distributions to future work.

The reader may wonder whether SNARGs of proximity [6] can be directly applied here to obtain a SNARG for distributions. For instance, can the verifier collect sufficiently many samples from its distribution, and then have the prover and verifier jointly run a SNARG of proximity on the empirical frequency vector obtained from these samples? We emphasize that this approach does not work, since the frequency vector obtained from samples is a poor estimate of the low-density part of the distribution. However, differences in the low-density part may be the main reason why two distributions are statistically far.

On the Necessity of Cryptography. A natural question is whether cryptography is necessary in our SNARG-Dist for NP (Theorem 1). In particular, does there exist a "MA-type" proof for general NP-properties over distributions with both sublinear proof and sublinear verifier's sample complexity? We observe that, even if we only require the proof size to be sublinear but allow the verifier to read the entire density function of the distribution, such a SNARG-Dist already implies a standard SNARG for NP. Thus, achieving sublinear proof sizes in SNARG-Dist for NP inherently requires cryptographic assumptions.

2.2 New Primitive: Sketching for Distribution Identity Testing

As a conceptual contribution, we identify an information-theoretical notion of *sketching for distribution identity testing*, which we will show, is essentially "equivalent" to SNARG-Dist.

At a high level, a sketching algorithm for distribution identity testing can compress a distribution D' over $[n]$ into a short sketch of size sublinear in n, and given a sublinear number of samples from any distribution D, one can use

the sketch to tell whether D and D' are the same, or they are ε-far in statistical distance. Specifically, a sketching scheme consists of a sketching algorithm Sketch and a tester algorithm Tester, defined as follows:

- Sketch: Given the full density function of a $1/\mathsf{poly}(n)$-grained distribution D' over $[n]$, the sketching algorithm is a randomized algorithm that "compresses" D' into a short sketch skt.
- Tester^D: Given a sketch skt and a set of samples from a distribution D, the testing algorithm can decide whether to accept or reject.

Correctness. We require that, for any D and D', Tester^D accepts a sketch of D with high probability if $D = D'$, and rejects with high probability if D and D' are ε-far.

We defer the formal definition of sketching for distributions to the full version. This notion can be viewed as between the identity testing for known distributions and for two unknown distributions. Indeed, the identity tester for unknown distributions directly [1] gives us a sketch of size $\tilde{O}_\varepsilon(n^{2/3})$ with Tester's sample complexity $O_\varepsilon(n^{2/3})$.

Equivalence to $\mathsf{SNARG\text{-}Dist}$. Sketching for distributions is a conceptually simpler primitive than $\mathsf{SNARG\text{-}Dist}$——it only provides the functionality of identity testing, and its correctness is defined by first fixing D and D', and then the randomness of Sketch and Tester are independently generated. This is in contrast to SNARG soundness, where the CRS is generated first, and then a malicious prover can depend on CRS in an arbitrary way. Nevertheless, our results show that sketching for identity testing is both *necessary and sufficient* for $\mathsf{SNARG\text{-}Dist}$, in the following sense.

Sufficiency. To obtain our positive result of non-adaptively sound $\mathsf{SNARG\text{-}Dist}$ for NP (Theorem 1), our construction of $\mathsf{SNARG\text{-}Dist}$ makes generic use of sketching for distribution identity testing. Moreover, the $\mathsf{SNARG\text{-}Dist}$ proof size and verifier's sample complexity are roughly the size of the sketch skt and Tester's sample complexity, respectively (ignoring a polynomial factor in security parameter λ). We defer an overview of our techniques to Sect. 3.1.

Necessity. To obtain our lower bound for $\mathsf{SNARG\text{-}Dist}$, indeed, our first step is to show that any $\mathsf{SNARG\text{-}Dist}$ for $\mathsf{P/poly}$ also implies sketching for distribution identity testing in the shared common randomness model, assuming hardness of LWE. Then we prove an information-theoretical lower bound for sketching.

We defer an overview of how SNARGs imply sketching to Sect. 3.3 and a formal theorem to the full version.

We believe this equivalence observation is of independent interest: to improve the proof size of our $\mathsf{SNARG\text{-}Dist}$ in Theorem 1, it suffices to improve the sketch size. On the other hand, to improve the lower bound of $\mathsf{SNARG\text{-}Dist}$ in Theorem 4, it suffices to improve our lower bound of sketching for distributions.

3 Technical Overview

We provide an overview of our techniques for constructing SNARG-Dist for NP (Theorem 1) in Sect. 3.1; and then an overview of our techniques for constructing SNARK-Dist for P (Theorem 2) in Sect. 3.2; and then give an overview of why SNARG-Dist imply sketching for identity testing in Sect. 3.3; and finally an overview of our lower bound techniques for sketching for distributions in Sect. 3.4.

3.1 SNARG for Distributions via Distributional Property Testing

Our starting point is the identity testing for two unknown distributions. Given sample access to two unknown distributions D and D' over the domain $[n]$, there exists a polynomial-time tester that only draws $\tilde{O}_\varepsilon(n^{2/3})$ samples from each distribution and can distinguish whether $D = D'$ or $\mathsf{SD}(D, D') > \varepsilon$ [1].

Initial Attempt. Using such a tester, our initial attempt to build SNARGs for distributions is as follows. The prover samples $\tilde{O}(n^{2/3})$ independent samples from its distribution D' and sends them to the verifier. The prover also attaches a SNARG for NP to prove that these samples are honestly generated from D' using the random coins, and that $D \in \mathcal{L}$. (For simplicity of this overview, we ignore the ρ-proximity gap between in our approximately decidable distribution property definition, and simply assume that for every $D \notin \mathcal{L}$, the decider of $\mathcal{L}$ rejects).

To apply a SNARG for NP here, we need an explicit sampling algorithm Sample that takes the density function of the distribution $D' = \{(i, D'[i])\}_{i\in[n]}$ and some random coins r as input, and outputs a sample from D'. This can be achieved by using the *inverse transformation sampling* in statistics [13]. We include this standard sampling algorithm in the full version. Since we assume prover's input D' is a $1/\mathsf{poly}(n)$-grained distribution, the random coins r we need in Sample are $O(\log \mathsf{poly}(n)) = O(\log n)$ bits per sample.

To prevent a cheating prover from choosing the random coins r maliciously, one may attempt to sample the random coins r during the trusted setup and put it in the CRS along with the CRS of SNARG for NP. To verify the proof, the verifier obtains $\tilde{O}(n^{2/3})$ samples from its distribution D, and it runs the tester first to ensure that D and D' are close, by using the samples of D' sent by the prover. Then the verifier runs the verifier of SNARG for NP to ensure D' is close to $\mathcal{L}$.

However, this construction does not work directly, since a cheating prover may first observe the randomness r in the CRS, and then maliciously select the distribution D^*. Then in the proof of soundness of this protocol, it is unclear how to apply the correctness of the property tester, since the correctness requires that the random coins are independent of the distributions D and D'.

Hash the Distribution. To resolve this issue, we first construct a 3-round public-coin argument, where we use a hash commitment to "fix" the distribution D' before sampling r, and then we will apply the Fiat-Shamir transformation to obtain a SNARG for distributions.

Specifically, we have the prover hash commit the distribution $\mathsf{h} = \mathsf{Hash}(\mathsf{hk}, D')$, and send the hash value h to the verifier in the first round. Recall that the $1/\mathsf{poly}(n)$-grained distribution D' is fully discrete and can be represented in $\tilde{O}(n)$ bits, and hence we can hash this bit string. Then we put the hash key hk in the CRS.

In the second round, the verifier sends a uniformly random string r as the random coins for the prover's sampler Sample, with which the prover generates $\tilde{O}(n^{2/3})$ samples from D'. Finally, the prover sends to the verifier a SNARG for NP proof for the statement

> "There exists an underlying distribution $D' \in \mathcal{L}$ and $\mathsf{h} = \mathsf{Hash}(\mathsf{hk}, D')$, and the samples are indeed obtained by the sampler $\mathsf{Sample}(D'; \mathsf{h})$ with randomness h."

Finally, the verifier runs the property tester and then verifies the proof of SNARGs for NP. The intuition for soundness is that the hash value h in the first round fixes some D' in a "malicious prover's mind", and hence the random coin r is sampled independently of D'. Then the correctness of property testing ensures that $\mathsf{SD}(D, D') < \varepsilon$. Finally, by the soundness of SNARG for NP, $D' \in \mathcal{L}$.

To make this 3-round public-coin protocol non-interactive, we apply the Fiat-Shamir transformation. For simplicity, we directly parse the hash value h as the random coins r of the sampler.

In summary, the protocol proceeds as follows:

- **Key Generation:** Sample a hash key hk and a CRS for a SNARG for NP.
- **Prover:** Given a distribution D' and its NP witness w, it does the following:
  - Hash the distribution $\mathsf{h} = \mathsf{Hash}(\mathsf{hk}, D')$ where $D' = \{(i, D'[i])\}_{i \in [n]}$ is represented in $\tilde{O}(n)$ bits since it is $1/\mathsf{poly}(n)$-grained.
  - Use the hash value h as the randomness to run the sampler $\mathsf{Sample}(D'; \mathsf{h})$ and get $\tilde{O}(n^{2/3})$ samples. Denote the samples as S.
  - Generate a SNARG for NP proof for the following statement:

    > "there exists an underlying distribution D' and a witness w such that: D' hashes to h, and S is the output of the sampler $\mathsf{Sample}(D'; \mathsf{h})$, and w certifies $D' \in \mathcal{L}$."

  - The proof consists of the hash value h, samples S, and the SNARG for NP proof.
- **Verifier:** Verifies the SNARG for NP proof using the hash value h and the samples S. It then draws another $\tilde{O}(n^{2/3})$ samples from its distribution D and runs the identity tester using S.

Applying Correlation Intractable Hash. The above construction can be proven secure in the random oracle model. However, our goal is to achieve a construction in the standard model, from standard cryptographic assumptions. Our next idea is to instantiate the hash function with a correlation intractable hash with somewhere statistical correlation intractability (SS-CI) [4]. Correlation intractable hash (CI hash) was originally proposed to instantiate Fiat-Shamir transformation [5]. It captures the following intuition about random functions: the input-output pair of a random function should not satisfy any correlations. CI hash is a keyed hash function that formalizes this as the following security requirement: for any *evasive* binary relation $R \subseteq \mathcal{X} \times \mathcal{Y}$, which means for any $x \in \mathcal{X}$, the fraction of y's such that $(x, y) \in R$ is only negligible in $\mathcal{Y}$. CI hash requires that for any efficient adversary, given the hash key hk, it cannot find an input x such that the input-output pair $(x, \mathsf{Hash}(\mathsf{hk}, x))$ satisfies the correlation R.

Somewhere statistical CI hash (SS-CI) is a CI hash with the following additional property: the hash key has another computationally indistinguishable mode hk_R, which can encode a binary relation R. Under the key hk_R, it is statistically impossible to find an input x such that $(x, \mathsf{Hash}(\mathsf{hk}_R, x)) \in R$. SS-CI hash was known from iO and LWE for efficiently verifiable product relations [22].

Given SS-CI hash, we modify the above construction as follows: replace the hash function with an SS-CI hash [22], then instantiate the SNARG for NP with an adaptively sound SNARG for NP from iO [40,41]. Now we obtain a SNARG for NP-distribution properties from standard assumptions.

Proving Non-adaptive Soundness. To prove the non-adaptive soundness of our construction, our high-level strategy is as follows. For any D that is ε-far from the language $\mathcal{L}$, we first define a "bad" binary relation BAD, which contains all pairs (D^*, h) such that D^* is ε-far from D but h makes the identity property tester accept with probability $> 1/3$. Namely,

$$\mathsf{BAD} = \{(D^*, \mathsf{h}) : \mathsf{SD}(D^*, D) \geq \varepsilon \text{ and } \Pr[\mathsf{Tester}^D(S) \text{ accepts}] > 1/3 \text{ where } S = \mathsf{Sample}(D^*; \mathsf{h})\}.$$

Then we switch SS-CI hash key to the mode $\mathsf{hk}_{\mathsf{BAD}}$. By SS-CI property, for any D^* that is ε-far from D, the randomness $\mathsf{h} = \mathsf{Hash}(\mathsf{hk}_{\mathsf{BAD}}, D^*)$ will not make the identity property tester Tester accept with probability $> 1/3$.

Given this property, we will further apply the soundness of SNARG for NP to prove the soundness of our construction. Our high-level strategy is as follows. Suppose a cheating prover causes the verifier to accept with high probability, let S be the samples sent by the cheating prover. By the above SS-CI property, information-theoretically there does not exist an underlying $D^* \in \mathcal{L}$ such that S is honestly computed by $\mathsf{Sample}(D^*; \mathsf{h})$, where h is the hash of D^*. Hence, the statement of the underlying SNARG for NP is always false, and the soundness of our construction follows from the adaptive-soundness of the underlying SNARG for NP.

However, there are two issues with the above proof sketch: (1) the relation BAD defined above is not necessarily evasive, but as we mentioned in the previous paragraph, CI hash is only known for evasive relations; (2) SS-CI hash is only known from efficiently verifiable relation [21,22], but our BAD is not efficiently verifiable—we do not know how to efficiently compute the acceptance probability of the tester, unless we open the "black box" of the identity property tester.

Solution for Issue 1: Parallel Repetition. To resolve the first issue, we repeat the protocol in parallel for $\ell = \Theta(\lambda)$ times. Namely, instead of viewing the output of Hash as a single h, we parse the output of Hash as a tuple of ℓ hash values $\mathsf{h}_1, \ldots, \mathsf{h}_\ell$. Then we have the prover generate ℓ sets of samples $S_1, \ldots, S_\ell$ where $S_i \leftarrow \mathsf{Sample}(D^*; \mathsf{h}_i)$. The verifier also runs $\mathsf{Tester}^D(S_i)$ for every $i \in [\ell]$. Now BAD becomes a "product relation" [21]:

$$\mathsf{BAD}' = \{(D^*, (\mathsf{h}_1, \ldots, \mathsf{h}_\ell)) : \mathsf{SD}(D^*, D) \geq \varepsilon \quad \text{and} \quad \text{for all } i \in [\ell], \Pr[\mathsf{Tester}^D(S_i) \text{ accepts}] > 1/3 \text{ where } S_i = \mathsf{Sample}(D^*; \mathsf{h}_i)\}.$$

Now we can argue that this BAD is an evasive relation as follows. We can assume that the correctness error of the identity property tester is $< 1/6$, i.e. $\mathbb{E}_{\mathsf{h}}[\Pr[\mathsf{Tester}^D(S \coloneqq \mathsf{Sample}(D^*; \mathsf{h})) \text{ accepts}]] < 1/6$ for any fixed D and D^*. By Markov's inequality, we have that for each repetition and every fixed D^*, there are at most 1/2-fraction of h_i's that are "bad" (i.e. $(D^*, \mathsf{h}_i) \in \mathsf{BAD}$). Since we have ℓ parallel repetitions, the fraction of $(\mathsf{h}_1, \ldots, \mathsf{h}_\ell)$'s that all "bad" is at most $(1/2)^\ell$ fraction. This proves that BAD' is an evasive relation.

Solution for Issue 2: Estimate the Accepting Probability. To resolve the second issue, our key observation is that, since we are proving non-adaptive soundness, D is already known during the reduction, and hence we can estimate the acceptance probability of Tester^D by randomly sampling a large set of samples $S'_1, \ldots, S'_{\ell'}$ from D, and hardcoding them into the description of bad relation. Namely, we run the identity property tester through all $S'_1, \ldots, S'_\ell$ and estimate the probability that the identity tester accepts. In the detailed proof, we will show that this estimation can be very close to the ground truth. Now we obtain an efficiently verifiable product relation $\widetilde{\mathsf{BAD}}_{S'_1,\ldots,S'_\ell}$, which is almost the same as BAD', with the only difference that, whenever we need $\Pr[\mathsf{Tester}^D(S_i) \text{ accepts}]$, we run Tester using $S'_1, \ldots, S_\ell$, and use the fraction of acceptance among all ℓ executions to estimate $\Pr[\mathsf{Tester}^D(S_i) \text{ accepts}]$. This gives us an efficiently verifiable product relation $\widetilde{\mathsf{BAD}}_{S'_1,\ldots,S'_\ell}$.

We note that the proof contains more details. For instance, we need to extend the SS-CI hash in [22] from efficiently projection-enumerable relations to efficiently verifiable product relations using the sub-sampling trick in [21]. We defer more details of the proofs to the full version.

3.2 SNARK for P-Distribution Properties from Simple Standard Assumptions

In this section, we overview the techniques for constructing SNARK-Dist for P from simpler standard assumptions, such as LWE (Theorem 2).

Initial Attempt: Applying RAM-SNARG. Our starting point is to replace the SNARG for NP in our previous SNARG-Dist for NP (Sect. 3.1) with a RAM-SNARG for P, since the latter was known from simple standard assumptions such as LWE, or subexponential DDH and QR, or subexponential DDH, or bilinear maps [9,11,24,29,31,39]. This eliminates the use of iO. Recall that RAM-SNARG for P allows a prover to convince a verifier that a polynomial-time RAM computation is correct, and the verifier only needs a Merkle hash of the input to verify the RAM-SNARG proof. The soundness of RAM-SNARG states that, as long as the Merkle hash provided to the verifier is honestly computed from the input, then no cheating prover can cause a verifier to accept an incorrect RAM computation result.

To apply RAM-SNARGs in our setting, specifically, we have the prover Merkle-hash the input distribution D to a hash root $\mathsf{rt} = \mathsf{MerkleHash}(D)$, and then replace the previous SNARGs for NP with a RAM-SNARG proving that, taking this D as input, output the correct Hash value h of D and the correct sample S from $\mathsf{Sample}(D; \mathsf{h})$. Then we additionally have the prover send the Merkle root rt to the verifier, so that the verifier can run RAM-SNARG verifier.

However, it is unclear how to prove the soundness of this approach, since the RAM-SNARG is only sound when rt is honestly computed from verifier's distribution D. But in our modified scheme above, the prover chooses rt, and hence a cheating prover may choose a wrong rt. To resolve this issue, we will have the prover further provide a proof that rt is correctly computed from some distribution. Then in the security reduction, we will *extract* a distribution D^* from any cheating proof and the verifier's distribution D, and prove that rt is the correct Merkle root of this D^* and that $\mathsf{SD}(D^*, D) \leq \varepsilon$.

Extraction via Syndrome Decoding. To enable such an extraction, our next idea is to apply the syndrome techniques used in the recent work, SNARGs of proximity [6]. In their setting, the statement is represented as a binary string, and they encode the *difference* between the statement "in prover's mind" and verifier's statement as an error-correcting-code syndrome of the statement. They then attach this syndrome as part of the proof. In their soundness analysis, they leverage syndrome decoding to extract the "statement in the prover's mind". However, it is unclear how to apply the same technique in our setting, because the statement itself is now a distribution, which is naturally not a binary string.

To resolve this issue, we embed distributions into binary strings, with the requirement that this embedding preserve statistical distance. Specifically, for each index $i \in [n]$, we can represent the real value $D[i]$ as a binary string of length $1/\mu$, where the h-th coordinate is a binary indicating whether $D[i] \leq h \cdot \mu$ for every $h \in [1/\mu]$. If we concatenate this representation for all $i \in [n]$, then

any μ-grained distribution D over $[n]$ can be represented as a binary string of length $n \cdot 1/\mu$. We denote this binary representation as $\langle D \rangle$. Then $\langle \cdot \rangle$ is a distance-preserving embedding from statistical distance (L_1-distance) to Hamming distance: for any μ-grained distribution D^* and D, $\langle D^* \rangle$ and $\langle D \rangle$ differ by $\mathsf{SD}(D^*, D)/\mu$ coordinates. This is because for each coordinate i, the representation of $D[i]$ and $D^*[i]$ differ by $|D[i] - D^*[i]|/\mu$ coordinates.

Given this embedding, if we can "encode" the Hamming error between $\langle D^* \rangle$ and $\langle D \rangle$ as part of the $\mathsf{SNARG\text{-}Dist}$ proof, then we can extract D^* from D and their Hamming error. This motivates us to leverage the syndrome decoding technique from coding theory. Recall that syndrome decoding enables a receiver to efficiently recover the original message from a corrupted version using a short auxiliary string called a syndrome. With appropriate codes, the syndrome length can be as short as the number of the corruptions——in our case, this is only $\tilde{O}(\varepsilon/\mu)$, which is sublinear for $\varepsilon = o(1)$ and $\mu \geq 1/n$.

To apply syndrome decoding, we have the prover additionally add a syndrome s of $\langle D \rangle$ to the $\mathsf{SNARG\text{-}Dist}$ proof, and we also use RAM-SNARG to prove that this syndrome is correctly computed from $\langle D \rangle$. In the soundness reduction, one may attempt to recover a $\langle D^* \rangle$ "in the prover's mind" via s and verifier's distribution $\langle D \rangle$ using syndrome decoding. However, we cannot even guarantee that the recovery algorithm succeeds, since the correctness of syndrome decoding needs to assume $\mathsf{SD}(D^*, D) < \varepsilon$ in the first place, and we do not have D^* before syndrome decoding.

Our next observation is that, if we have *argument of knowledge (AoK)* property, then this reduction strategy works——we first use AoK to extract $\langle D^* \rangle$ and the RAM-SNARG ensures that s is the correct syndrome of this $\langle D^* \rangle$, then the syndrome recovery algorithm must output the same $\langle D^* \rangle$ since D^* and D are ε-close guaranteed by the property tester. However, obtaining such a *global* extracted witness is essentially asking for a SNARK for an NP statement about the entire distribution, which is difficult to achieve from standard assumptions.

Divide and BARG. To avoid a global extraction, we apply *batch arguments (BARGs)*, which enables *partial* witness extraction. Recall that a BARG for NP enables a prover to convince a verifier that a batch of instances $x_1, \ldots, x_k$ are all true via a single proof that is shorter than the total witness combined. For security, BARG provides *somewhere extraction* property, which enables an extractor to recover a valid witness for *one* of the statements, given any accepting BARG proof.

To apply BARGs, our high-level strategy is to have the prover divide the distribution D into k blocks, and for each block, we compute the syndrome, the CI hash value, and samples from Sample, and then use a BARG to prove that this computation is correct for all blocks. Finally, the prover Merkle-hashes the entire distribution D and uses RAM-SNARG for P to prove $D \in \mathcal{L}$.

Specifically, we divide D by dividing the domain $[n]$ into $k = n^\delta$ blocks in a natural way, and obtain k distributions $\bar{D_1}, \ldots, \bar{D_k}$ by restricting the density function of D on each block. To ensure each $\bar{D_i}$ is still a distribution with density mass summed to 1, we introduce an additional dummy element e_i and assign e_i with the remaining probability mass in D_i. Namely,

$$D_i(e_i) = 1 - \sum_{x \in \mathrm{supp}(\bar{D}_i)} \bar{D}_i(x).$$

Now we obtain a distribution D_i for each i.

Summary of the Construction. Putting the above components together, we now arrive at a construction for $\mathsf{SNARK\text{-}Dist}$ as follows. We highlight a few new technical challenges and our resolutions in red color.

- **Key Generation:** Generate a CI hash key hk and CRSes for the BARG and RAM-SNARG.
- **Prover:** Given a distribution D, the prover proceeds as follows:
  - Partition the domain into $k = n^\delta$ blocks and construct the corresponding distributions $D_1, \ldots, D_k$ by adding dummy elements as described above.
  - For each block $i \in [k]$:
    - Compute a hash value $\mathsf{h}_i \leftarrow \mathsf{Hash}(\mathsf{hk}, D_i)$.
    - Use h_i as randomness to sample from D_i: $\mathsf{skt}_i := \mathsf{Sample}(D_i; h_i)$.
    - Compute a syndrome s_i of $\langle D_i \rangle$.
  - Compute a Merkle hash of D, and denote the Merkle root as rt.
  - Produce a BARG proving that h_i, skt_i, s_i are correctly computed from some underlying distribution D_i as a witness. Moreover, to ensure each witness D_i is consistent with the Merkle hash rt of the "global" D, the BARG also proves that for every i, there exists a local opening (root-to-leaf path in the Merkle hash tree) of rt for the i-th block D_i.
  - Produce a RAM-SNARG proving that $D \in \mathcal{L}$.
  - Send $(\mathsf{h}_i, \mathsf{skt}_i, s_i)$ for all blocks $i \in [k]$, the root rt, and the RAM-SNARG proofs to the verifier.
- **Verifier:** Using the received proof and a sampling oracle for D, the verifier does the following:
  - Verify each skt_i using the identity tester Tester on the distribution D_i for the i-th block.
    However, since the verifier does not have direct access to D_i, we need to simulate samples from D_i by sampling from D. To simulate this, each time we draw a sample from D, if it falls in the i-th block, then we treat it as a sample from D_i. To account for the aforementioned dummy element e_i, if the sample from D falls outside the i-block, we assign that sample to e_i.
  - Verify the BARG proof and the RAM-SNARG proof.

Proving Argument of Knowledge. To prove the non-adaptive soundness of the above construction, we indeed prove a stronger property —— non-adaptive argument of knowledge, which states that given any verifier's distribution D and any accepting proof, an extractor can efficiently extract a distribution D^* that is ε-close to D and $D^* \in \mathcal{L}$.

Extractor Construction. Our knowledge extractor takes the verifier's distribution D and any accepting proof as its input, divides D into blocks $D_i (i \in [k])$, then runs syndrome decoding using s_i in the accepting proof with $\langle D_i \rangle$ to recover a $\langle D_i^* \rangle$ for every $i \in [k]$. Finally, it combines all "local" D_i^*'s to form a "global" distribution D^*. Next, we will prove that $\mathsf{SD}(D^*, D) \leq \varepsilon$ and $D^* \in \mathcal{L}$.

Proving $\mathsf{SD}(D^*, D) \leq \varepsilon$ *via BARG.* Indeed, it is not even immediately clear that the syndrome decoding algorithm can successfully output some D_i^*, but we will prove that this must be the case and that D_i^* is very close to D_i using BARG soundness. For every index i, we leverage the somewhere extractability of BARGs to argue that, whenever we set the BARG CRS to encode this i, we can extract from the BARG proof a distribution D_i^+ such that s_i is the correctly computed syndrome of $\langle D_i^+ \rangle$. Moreover, by the CI hash security and the correctness of the identity property tester, D_i^+ is close to D_i, and therefore $\langle D_i^+ \rangle$ is close to $\langle D_i \rangle$ in Hamming distance. Now, since the string $\langle D_i^+ \rangle$ underlying the syndrome s_i always exists, we can apply the correctness of syndrome decoding to argue $D_i^+ = D_i^*$. This strategy works for any specific i, and hence we can switch the CRS of the BARG for every i and prove that syndrome decoding of D_i^* succeeds, and D_i^* is close to D_i.

Proving $D^* \in \mathcal{L}$ *via RAM-SNARG.* To prove $D^* \in \mathcal{L}$, we will leverage RAM-SNARG soundness. Suppose we can argue that the Merkle hash of the extracted D^* is the same as rt sent by the cheating prover, then we can conclude $D^* \in \mathcal{L}$ by applying RAM-SNARG soundness directly. Hence, it remains to prove

$$\mathsf{MerkleHash}(D^*) = \mathsf{rt}.$$

We prove this using the no-signaling property of the BARG extraction: we will prove consistency incrementally along the computation of Merkle hash root rt. Specifically, we view the computation of the Merkle hash root as an incremental process on an evolving forest. We start from the leaves and add them in order. Whenever we add a new leaf, we form the trees from all leaves we added so far "as much as we can". We illustrate this step-by-step evolution from leaves to the final root in Fig. 1.

Let R_i denote the set of roots of those trees when the first i leaves are added. We will inductively prove that, if we use the BARG to extract R_i in the accepting proof, then the BARG extracted hash values in R_i are the same as the values of R_i computed from D^*.

At each induction step, when the i-th leaf node is added in the picture, we use BARG to extract the root-to-leaf path for this leaf as well as the old roots R_{i-1}. By the no-signaling property of BARG extraction, R_{i-1} is still consistent with the Merkle hash tree of D^*. Moreover, the newly extracted root-to-leaf path for i must be consistent with R_{i-1} due to collision resistance, and the i-th leaf D_i^+ extracted from BARG must be consistent with D_i^* following our argument in the previous paragraph. Therefore, we can merge them and argue that R_i is also consistent with the Merkle hash tree of D^*. This finishes the induction step. In the last step of the induction, we conclude that R_n is consistent with the Merkle

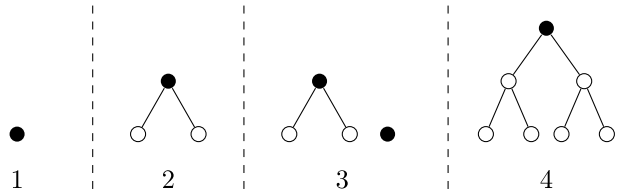


Fig. 1. Step-by-step computation of the Merkle hash root viewed as an evolving forest. The hash is computed incrementally from the leaves to the root: starting from individual leaf nodes, intermediate parent nodes are derived by hashing their children as soon as we can. We denote by R_i the set of roots of the forest formed by the first i leaf nodes, which are the black nodes in the figure.

hash root of D^* while R_n must also be rt. This ensures $\mathsf{MerkleHash}(D^*) = \mathsf{rt}$ and finishes the proof.

New Abstraction: BARG for Distributions. We observe that implicit in our construction above is a proof system for a batch of distributions $D_1, \ldots, D_k$. Hence, we abstract a new notion of *batch argument for distributions* (BARG-Dist) for NP, which may be of independent interest. We first construct BARG-Dist for NP, and then use it generically to build SNARK-Dist for P.

3.3 SNARGs for Distributions Imply Sketching for Identity Testing

In this section, we show that any SNARG-Dist for P/poly with proof length $\ell = \ell(\lambda)$ can be turned into a sketch for identity testing of length $\ell + \mathsf{poly}(\lambda)$ in the shared common randomness model. This shows the necessity of sketching for SNARG-Dist, and is an important step in proving the lower bound of SNARG-Dist.

Suppose we have a SNARG for distribution, a naive approach to constructing a sketching scheme is the following:

- $\mathsf{Sketch}(D)$: Run the prover algorithm of SNARGs for distributions for the language $\mathcal{L} = \{D\}$ to generate a SNARG proof π, and output this proof as the sketch.
- $\mathsf{Tester}^{D'}$: Given a sketch π (i.e. a proof of SNARG for distribution), run the SNARG verifier on the same language $\mathcal{L}$ to verify π, and output whatever the SNARG verifier outputs.

Our hope is that when $D' = D$, π can be accepted by the completeness of SNARGs for distributions, and when D' and D are ε-far, then π would be rejected by the soundness of SNARGs for distributions. However, this natural strategy fails because Tester also needs to know the full description of the language $\mathcal{L}$, which is D itself. To enable this, one may attempt to include D as part of the sketch. But if we use this approach, we will finally arrive at a trivial communication complexity lower bound. That is, the identity property testing lower

bound for known distributions: with full description of D, Bob needs $\Omega(\sqrt{n})$ samples to do identity testing.

To address this issue, we apply a hash function to D to compress the full description of D to a short hash commitment. Now we model this problem as a $\mathsf{P/poly}$ distribution property:

$$\mathcal{L}_{\mathsf{hk},c} = \{D \mid \mathsf{Hash}(\mathsf{hk}_\lambda, D) = c_\lambda\},$$

where $\{(\mathsf{hk}_\lambda, c_\lambda)\}_{\lambda\in\mathbb{N}}$ is a sequence of polynomial-size advice. The reader may wonder how $\mathsf{SNARG\text{-}Dist}$ for $\mathsf{P/poly}$ is defined. Since the prover and the verifier need to know the advice, we allow the prover and verifier in $\mathsf{SNARG\text{-}Dist}$ for $\mathsf{P/poly}$ to take the advice as input, but still require them to be *uniform* polynomial-time algorithms. Indeed, most of the concretely efficient classical SNARGs are for $\mathsf{NP/poly}$ in this sense.

Now, since the length of hash commitment c is short, we can have Sketch directly attach it in the short sketch. This leads to the following modified reduction from SNARGs for distributions to sketching scheme:

- $\mathsf{Sketch}(D)$: Hash the distribution D to obtain a short hash commitment c, where the hash key hk is obtained from shared common randomness. Then we run the SNARG prover algorithm to generate the proof π. The sketch consists of both c and π.
- $\mathsf{Tester}^{D'}$: Given the sketch (c, π), it runs the SNARG verifier on the $\mathsf{P/poly}$ language $\mathcal{L}$ with the advice hk, c, where hk is obtained from the shared common randomness.

When proving the correctness of our reduction, we first imagine an ideal world, where we have a perfect collision-resistant hash function Hash. Then the preimage of $\mathsf{Hash}(\mathsf{hk}, \cdot) = c$ can only be D. Then the correctness of Tester follows from the completeness and soundness of the $\mathsf{SNARG\text{-}Dist}$. However, since we require that c is short in length, information theoretically, the hash function always has collisions. Then it is unclear how to prove the correctness of the above reduction.

Single-Point Perfect Binding Hash from FHE. To address this issue, we construct a hash function that has two indistinguishable modes (similar to somewhere statistical binding hash [23]): a real-world mode $\mathsf{hk}_{\mathsf{real}}$ and a binding mode $\mathsf{hk}_{\mathsf{fake}}$. Under the binding mode key $\mathsf{hk}_{\mathsf{fake}}$, the hash function is "single-point perfect binding", which means $\mathsf{hk}_{\mathsf{fake}}$ encodes the full information of some distribution D^*, and information-theoretically no other strings will be hashed to $\mathsf{Hash}(\mathsf{hk}_{\mathsf{fake}}, D^*)$

When proving the correctness of sketching algorithm above, we first fix D and D^*, then we shift from the real-world to the binding mode since they are indistinguishable. Now, under the binding mode key $\mathsf{hk}_{\mathsf{fake}}$, which encodes D^*, no other distributions hash to the same $\mathsf{Hash}(\mathsf{hk}_{\mathsf{fake}}, D^*)$. Finally, the correctness of the reduction can be derived from the completeness and soundness of $\mathsf{SNARG\text{-}Dist}$.

We construct Hash with the above properties from leveled FHE, as follows. The real-world mode key $\mathsf{hk}_{\mathsf{real}}$ consists of a bunch of pseudorandom FHE ciphertexts, while the binding mode key $\mathsf{hk}_{\mathsf{fake}}$ for D^* consists of FHE encryption of D^*. To hash any distribution D, we homomorphically run the equality checking function over $\mathsf{hk}_{\mathsf{fake}}$, and use the evaluated ciphertext as the hash commitment c. Now, only D^* is hashed to FHE ciphertexts of 1 under $\mathsf{hk}_{\mathsf{fake}}$, and all other $D \neq D^*$ are hashed to FHE ciphertexts of 0.

We defer more details to the technical sections in the full version.

3.4 Lower Bound of Sketching for Distribution Identity Testing

In this section, we give an overview of our techniques for proving a lower bound of sketching for distribution identity testing:

Theorem 5 (Sketching lower bound, Informal Version). *For any constant $\varepsilon > 0$, any integer n, and any real $\delta \in (1/2, 2/3)$, the sketch size for any sketching for distribution testing is at least $\Omega(n^{3-4\delta})$ if the* Tester *sample complexity is $O(n^{\delta})$.*

To prove this lower bound of sketching, we first construct a pair of "hard" distributions D_0 and D_α, where $\alpha \in (0, 1)$ is a real number. We will prove that, even given a small sketch of D_0, the tester $\mathsf{Tester}^{D_\alpha}$ given samples of D_α cannot distinguish between the case $\alpha = 0$ and the case $\alpha > \varepsilon$ for some constant ε. Therefore, we can prove Theorem 5. Our hard distributions are inspired by the lower bound of identity distribution testing [1], but with significant modifications to prove lower bounds for sketching schemes.

Constructing Hard Distributions. D_0 and D_α each consist of two parts: the *heavy part* H consisting of the elements that have high densities, and the *light part* L consisting of the elements that have low densities. For D_0, the heavy part contains $n^{2-2\delta}$ elements with high density $1/2n^{2-2\delta}$ and n elements with low density $1/2n$. D_α has the same heavy part as D_0, but its light part is a "perturbed" version of the light part of D_0. The perturbation is specified by a partition of the light set $L = L^+ \cup L^0 \cup L^-$—the density in L^+ is increased by $1/2n$, and the density in L^0 remains unchanged, and the density in L^- is decreased by $1/2n$. Since D_α needs to be a distribution, we require $|L^+| = |L^-|$ so that its density function sums to 1, and we use $\alpha = |L^+|/n$ to parameterize how much we perturb D_0. In particular, when $\alpha = 0$, then D_α is the same as D_0. An illustration of D_0 and D_α is given in Fig. 2.

We now proceed to prove that, even given a short sketch of D_0, the tester $\mathsf{Tester}^{D_\alpha}$ cannot distinguish between the case $\alpha = 0$ and $\alpha > \varepsilon$. To show this, we first observe that the only information $\mathsf{Tester}^{D_\alpha}$ can obtain from D_α is the *frequency* of the samples from D_α, apart from the sketch skt. This is because the samples are i.i.d., and hence they are all symmetric. Recall that, for every element $i \in [2n]$, the frequency of i is the number of times that i appears in the samples. Next, we analyze the behavior of the frequency for D_α.

Analyzing the Frequency of Samples from D_α. Suppose the tester Tester draws n^δ samples from D_α, where $\delta \in (1/2, 2/3)$. Then the frequency for every light part element is at most 2 with high probability, since the probability of having a 3-way collision among the light part is bounded by $(1/n)^2 \cdot \binom{n^\delta}{3} < 1$ for $\delta < 2/3$.

Now we can only have frequencies $0, 1, 2$ in the light part. Among them, we are particularly interested in the elements with frequency 2, i.e. 2-way collisions. Intuitively, estimating the number of 2-way collisions is the key to uniformity testing, as shown in previous work [1,36]. In our setting, distinguishing D_0 and $D_\alpha(\alpha > \varepsilon)$ is essentially a uniformity testing restricted only to the light part. This is because D_0 and D_α share the same heavy part and the light part of D_0 is uniform. We formalize this intuition as follows.

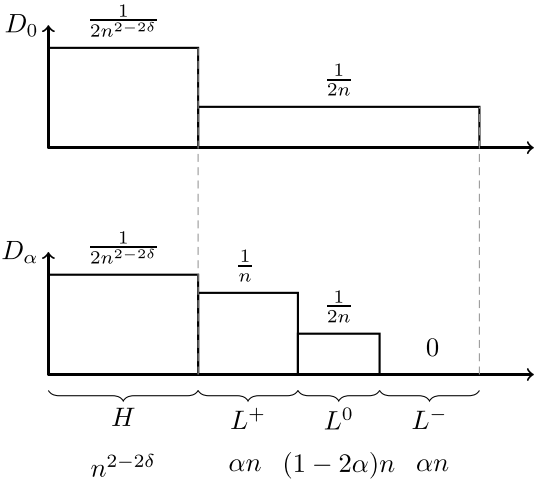


Fig. 2. Illustration of distributions D_0 and D_α. D_0 and D_α share the same heavy part H and light part L, but the light part of D_α is perturbed in L^+ and L^-.

Let $S_2^L \subseteq L$ be the set of elements that have frequency 2 in the light part among the n^δ samples of D_α. We prove that the expectation of $|S_2^L|$ is roughly $(1/8 + \alpha/4) \cdot n^{2\delta-1}$, which scales linearly in α and is of magnitude $O(n^{2\delta-1})$. Moreover, the standard deviation of $|S_2^L|$ is of magnitude $O(\sqrt{n^{2\delta-1}})$, which is much smaller than $|S_2^L|$ itself. Hence, if one can count $|S_2^L|$, then it can well-estimate α and hence construct Tester.

Intuition of Hardness for Distinguishing D_0 and D_α. Our intuition for the lower bound is that the opposite direction is also true: constructing a tester Tester is as hard as estimating $|S_2^L|$. This is because if $\mathsf{Tester}^{D_\alpha}$ correctly distinguishes $\alpha = 0$ and $\alpha > \varepsilon$, then its decision gives us an estimation of α and hence also a good estimation of $|S_2^L|$.

However, $\mathsf{Tester}^{D_\alpha}$ does not know which indices in $[2n]$ belong to the heavy or light part. As a result, Tester cannot directly count the 2-way collisions in the

light part. Instead, the tester can only observe the total 2-way collision count among all samples, which includes 2-way collisions in the light part as well as the 2-way collisions in the heavy part. However, note that the standard deviation of 2-way collisions in the heavy part is $\Theta(n^{2\delta-1})$, which can be viewed as a large "noise" that smudges $|S_2^L|$, since this standard deviation is as large as the value of $|S_2^L|$ itself even for the largest $\alpha = 1/2$.

This is the key intuition behind our lower bound proof. However, so far this intuition has not taken the sketch skt into account—what if, given the sketch skt, Tester can well-estimate $|S_2^L|$? Intuitively, this is still unlikely, since to encode the full information of the heavy part (or the light part) in the sketch, one needs the sketch size to be at least $|H| = \Omega(n^{2-2\delta})$. To formalize this intuition, we will abstract this problem as a communication complexity problem.

Core of the Sketching Problem: Promise Set Difference The above intuition motivates us to abstract the core of the sketching for distributions as the following communication complexity problem between two parties, Alice and Bob.

- **Alice:** This is the party who runs $\mathsf{Sketch}(D_0)$. It has full knowledge of the distribution D_0 and therefore knows exactly which indices belong to the heavy part.
- **Bob:** This is the party that runs $\mathsf{Tester}^{D_\alpha}$. It only observes the frequency from samples and thus only sees a bunch of 0's, 1's, and 2's. (Note that Bob can also see some 3's, 4's, ..., but those frequencies are not useful because they must come from the heavy part.) Bob does not know whether those 2's are from the heavy part or the light part of his distribution.
- **Goal:** The problem is how to design a one-way communication protocol from Alice to Bob that lets Bob estimate the count of 2's restricted to the light part.

This problem is essentially a communication complexity problem for set difference: Alice knows one set $X \subseteq [2n]$, which represents the set of heavy part indices H, while Bob holds another set $Y \subseteq [2n]$, which represents the set of indices with frequency of 2 when sampling from D_α. The goal is to let Bob estimate $|Y \backslash X|$ within a constant multiplicative factor.

This resembles the classical set difference problem. However, our instance is not arbitrary: the sizes of X and Y are close to their expectations of heavy part size and 2-way collisions, respectively. Also recall that $|Y \backslash X|$ represents $|S_2^L|$, whose expectation is of size $(1/8 + \alpha/4) \cdot n^{2\delta-1}$ where $\alpha \in [0, 1/2]$. Hence, we have a "promise": $n^{2\delta-1}/8 \leq |Y \backslash X| \leq n^{2\delta-1}/4$.

We further simplify the problem by removing the expected number of 2-way collisions in the heavy part. The reason why we can do this is by the discussion before: the noise in the number of heavy part 2-way collisions hides information about α in light part 2-way collisions $|S_2^L|$. Therefore, when defining the instance, we let Y include all light collisions together with some of the heavy collisions on the order of the standard deviation of heavy part 2-way collisions.

Promise Set Difference. In summary, we define our *Promise Set Difference* problem as follows.

- **Alice:** has a set X with promise $|X| = \Theta(n^{2-2\delta})$.
- **Bob:** has a set Y with promise $|Y| = n^{2\delta-1}/4$ and $|Y \backslash X|$ has promise $n^{2\delta-1}/8 \leq |Y \backslash X| \leq n^{2\delta-1}/4$.
- **Goal:** Alice can send a one-round message to Bob. The goal is to have Bob decide whether $|Y \backslash X| = n^{2\delta-1}/8$ or $|Y \backslash X| > (1/8 + \alpha/4) \cdot n^{2\delta-1}$ for some constant α.

We defer a formal version of our definition of promise set difference in the full version.

Reducing Promise Set Difference to Sketching. The analysis above only gives us an intuition that the promise set difference problem can be reduced to the sketching problem. To make this argument formal, we need to show an explicit reduction from the promise set difference problem to the sketching problem. Following our intuition above, one may attempt to construct the reduction as follows.

- **Alice:** Given a subset X, Alice constructs D_0 by marking elements in X as heavy part and treating all other elements in the universe as light part. Then it runs $\mathsf{Sketch}(D_0)$ and sends the sketch to Bob.
- **Bob:** Given a subset Y, Bob constructs a set of samples $\tilde{D_\alpha}$ to "simulate" the real samples from D_α (for some α to be determined). Then it will run Tester using $\tilde{D_\alpha}$ and the sketch sent by Alice.
  To construct $\tilde{D_\alpha}$, Bob assigns frequency 2 to every element in Y, since we view the set Y as the set of elements that have frequency 2 in the above discussion.

Choosing the Parameter α. Intuitively, this reduction is correct because the number of light part 2-way collisions in $\tilde{D_\alpha}$ is now exactly $|Y \backslash X|$. Then by choosing α such that $|Y \backslash X| = (1/8 + \alpha/4) \cdot n^{2\delta-1}$, the sketching scheme can estimate α, and therefore also estimate $|Y \backslash X|$. This solves the original promise set difference problem. The reader may wonder if we can really let α depend on both X and Y in the reduction. Indeed, this is not a problem since the reduction algorithm (Alice and Bob) does not need to know α explicitly——α is only used in the proof of correctness.

To make the correctness proof go through, we need to prove that $\tilde{D_\alpha}$ is close to the real samples from D_α. However, so far in the simulated samples $\tilde{D_\alpha}$, there are only elements with frequency 2's, while in the real samples from D_α, there could also be elements with frequency 1 or frequency ≥ 3. Hence, to simulate a real set of samples from D_α, Bob also needs to add those frequencies in $\tilde{D_\alpha}$.

Adding Frequency of 1. Intuitively, the number of frequency 1 Bob added to D'_α should be the expectation of the number of frequency 1 in the real samples from D_α, and this expectation is roughly n^δ. A naive attempt to add those 1's is to have Bob randomly sample n^δ elements and assign their frequencies as 1's. However, this does not work, since the frequency of 1's is not evenly distributed among the heavy part and the light part——indeed, the heavy and the light part have an equal number of frequency of 1's in expectation, but the size of the heavy part is only $n^{2-2\delta}$ while the size of the light part is much larger (linear in n).

We resolve this issue by extending the universe with new elements, and we have Alice and Bob act collaboratively to add those frequencies of 1's. Specifically, suppose the samples and distributions are originally over some finite set universe U. Now, we extend U to $U \cup H_1 \cup L_1$ with $|H_1| = |L_1| = n^\delta/2$, where H_1 represents the set of elements whose frequency is 1 in the heavy part, and L_1 represents the set of elements whose frequency is 1 in the light part.

To do this extension, we have Alice insert elements of H_1 into the heavy part and elements of L_1 into the light part of D_0, while Bob assigns frequency 1 to all elements in $H_1 \cup L_1$ in $\tilde{D_\alpha}$. Now we can ensure both the heavy part and the light part have exactly n^δ number of elements with frequency 1's.

Adding Frequency of ≥ 3. Following our analysis on the frequency of samples from D_α, frequencies ≥ 3 appear only in the heavy part. Furthermore, we only need to consider frequencies up to $O(\log n)$, since we can prove that the expected number of indices with frequency $\omega(\log n)$ is $o(1)$.

To add those frequencies to $\tilde{D_\alpha}$, we further extend the universe with sets $H_3, H_4, \ldots, H_{O(\log n)}$, where each H_i has size equal to the expected number of heavy elements with frequency i. Then Alice inserts elements of each H_i into the heavy part of D_0, and Bob assigns frequency i to these elements in $\tilde{D_\alpha}$. In this way, we can ensure those added frequencies only appear in the heavy part, and their number equals the expectation of the number of those frequencies in the real samples from D_α.

Randomly Shuffling the Universe. So far, the simulated samples $\tilde{D_\alpha}$ contain the correct number of elements for each frequency. However, $\tilde{D_\alpha}$ still does not behave like a real set of samples from D_α. For example, all 3-way collisions will be on H_3, while in the real samples from D_α, 3-way collisions are randomly distributed in the heavy part.

To resolve this issue, we further have Alice and Bob use shared randomness to randomly permute the universe before running the sketching scheme in our reduction. This ensures that the distribution of $\tilde{D_\alpha}$ is close to the real samples from D_α.

Proving Correctness of the Reduction. To prove the correctness of the reduction, we will first prove that the joint distribution of $(D_0, \tilde{D_\alpha})$ produced by the reduction is "close" to the joint distribution obtained when $\tilde{D_\alpha}$ is sampled from the real distribution D_α. Then, by the correctness of the sketching algorithm, when

$\tilde{D_\alpha}$ is a real sample from D_α, the tester can well-estimate the value of α, which corresponds to $|Y \backslash X|$ in our construction. This finishes the proof of correctness.

A naive approach to formalize the closeness between the simulated $(D_0, \tilde{D_\alpha})$ and the real samples from D_α, is to bound the statistical distance between them. However, this statistical distance is close to 1 in our setting and therefore does not yield a meaningful lower bound. To obtain a tighter bound, we instead analyze the Rényi divergence between the two distributions. We show that the Rényi divergence of infinite order R_∞ between the real sampling process and our reduction is bounded by a polynomial of n.

Then we apply the following *probability preservation property* of Rényi divergence: for any distribution D_1 and D_2, and for any event BAD, we have $D_1(\mathsf{BAD}) \leq D_2(\mathsf{BAD}) \cdot R_\infty(D_1 \| D_2)$. We let BAD denote the event that the sketching algorithm fails to decide correctly. Then by the above probability preservation property, if the sketching algorithm decides correctly over the samples from real distributions D_α except with a small probability $D_2(\mathsf{BAD})$, then the same sketching algorithm still decides correctly over the simulated samples given by our reduction. This would complete the correctness proof.

However, if the error probability of the sketching algorithm is a constant, then Rényi divergence does not give us any non-trivial bound on the correctness of the reduction. Hence, we additionally amplify the correctness probability using parallel repetition in the last step of the reduction. This repetition reduces the overall BAD probability to $1/\mathsf{poly}(n)$ and completes the correctness proof of the reduction. We defer more details to the full version.

Lower Bound for Promise Set Difference via Index Problem. Finally, we prove the lower bound of our promise set difference by reducing from the Index Problem in communication complexity. Recall that Index Problem is the following problem: One party Alice has a binary vector $x \in \{0,1\}^m$ and the other party Bob has an index $i \in [m]$. The goal is to let Bob learn x_i. The communication complexity lower bound of the index problem is well-known to be $\Omega(m)$.

To reduce from Index Problem, our key observation is that, if we divide every parameter in promise set difference problem (See Sect. 3.4) by $n^{2\delta-1}/4$, then the promise set difference problem becomes the following problem. The universe size becomes $n^{3-4\delta}$, $|X| = \Theta(n^{3-4\delta})$, and $|Y| = 1$. Now we can view this version of the promise set difference problem as an Index Problem, in the following way. Y can be viewed as an index i, and X can be viewed as a binary vector $\mathbf{x}$ whose coordinates of 1's happen to be X. Then $|Y \backslash X|$ is either 0 or 1, corresponding to the i-th coordinate of the vector $\mathbf{x}$.

Based on this observation, we construct the reduction to promise set difference problem from index problem by "repeating in parallel" an index problem for $n^{2\delta-1}$-times, as follows.

- **Alice:** Given a vector $\mathbf{x} \in \{0,1\}^{n^{3-4\delta}}$ from Index Problem, Alice converts $\mathbf{x}$ to a set $X \subseteq [n^{3-4\delta}]$ which contains all indices that are 1's in $\mathbf{x}$. It then

"parallelly copies" the current universe $U = [n^{3-4\delta}]$ for $(\ell = n^{2\delta-1})$-times to obtain $U_1, \ldots, U_\ell$, where each U_i is a fresh copy of U.
Then Alice also copies X to $X_1 \subseteq U_1, \ldots, X_\ell \subseteq U_\ell$ accordingly (by identifying the elements of U_i with U in any natural way). After the repetition, the universe now becomes $U_1 \cup U_2 \cup \ldots \cup U_\ell$.

- **Bob:** Given an index $i \in [n^{3-4\delta}]$, it constructs $Y = \{i\}$, and also "parallelly copies" it ℓ-times to obtain $Y_1 \subseteq U_1, \ldots, Y_\ell \subseteq U_\ell$.
- Finally, Alice and Bob jointly run the Promise Set Difference protocol (See Sect. 3.4) on the input $X_1 \cup X_2 \cup \ldots \cup X_\ell$ and $Y_1 \cup Y_2 \cup \ldots \cup Y_\ell$ respectively to solve the Index Problem.

From this reduction, we can argue that if the Promise Set Difference problem has communication complexity $o(n^{3-4\delta})$, then we can also have a one-way communication protocol for Index Problem with sublinear communication, contradicting the well-known lower bound for Index Problem. This gives us the lower bound for the Promise Set Difference problem.

We defer the details of the proof to the full version.

4 Sketch for Distribution Testing

Syntax. A Sketch for Distribution Testing consists of the following algorithms:

$\mathsf{Sketch}(D) \to \mathsf{skt}$. This is a probabilistic polynomial-time algorithm that takes as input a distribution D. It outputs a sketch of the distribution skt.
$\mathsf{Tester}^D(\mathsf{skt}) \to 0/1$. This probabilistic polynomial-time oracle algorithm takes as input samples of distribution D. It outputs a bit (1 to accept, 0 to reject).

Definition 1. *A Sketch for Distribution Testing is required to satisfy the following properties:*

Completeness. *For any distribution D,*

$$\Pr\left[\mathsf{Tester}^D(\mathsf{skt}) = 1 : \mathsf{skt} \leftarrow \mathsf{Sketch}(D)\right] \geq 2/3.$$

Soundness. *For any distributions D, D^* where $|D - D^*| > \varepsilon$, we have*

$$\Pr\left[\mathsf{Tester}^D(\mathsf{skt}) = 1 : \mathsf{skt} \leftarrow \mathsf{Sketch}(D^*)\right] < 1/3.$$

To construct a succinct distribution sketching scheme, we can leverage the foundational work on distribution testing, specifically the problem of determining if two unknown distributions are close or far from one another. In particular, this task can be achieved with a sublinear number of samples [1]. This insight forms the basis for a simple construction of a sketching scheme.

Theorem 6 (Efficient Equality Testing for Distributions [1]). *There exists an unconditional probabilistic polynomial-time algorithm* Equal *with distance ε that can test whether two distributions D, D^* are equal up to ε additive error with sublinear samples. The algorithm satisfies the following properties:*

Correctness. For any probability distributions D over a discrete domain $\Omega_n = [n]$,

$$\Pr\left[\mathsf{Equal}^{D,D}() = 1\right] \geq \frac{2}{3}.$$

Soundness. For any two probability distributions D, D^* over a discrete domain $\Omega_n = [n]$ such that $|D - D^*| \geq \varepsilon$,

$$\Pr\left[\mathsf{Equal}^{D,D^*}() = 1\right] \leq \frac{1}{3}.$$

Efficiency. Additionally, the algorithm Equal^{D,D^*} also has $\tilde{O}(n^{2/3})$ sample complexity for both oracles. More exactly, it makes $O(n^{2/3}\varepsilon^{-4}\log n)$ oracle calls to sample from D and $O(n^{2/3}\varepsilon^{-4}\log n)$ oracle calls to sample from D^*.

Theorem 7 (Succinct Distribution Sketching Scheme). *There exists a succinct distribution sketching scheme with $O(n^{2/3}\varepsilon^{-4}\log n)$ sketch complexity and $O(n^{2/3}\varepsilon^{-4}\log n)$ sample complexity.*

The construction of the scheme $(\mathsf{Sketch}, \mathsf{Tester})$ works as follows:

Sketching Phase. Sketch takes D as input and draws $m = O(n^{2/3}\varepsilon^{-4}\log n)$ independent samples $x_1, x_2, \ldots, x_m$ from D. These samples are then compiled into the sketch $\mathsf{skt} = (x_1, x_2, \ldots, x_m)$.

Testing Phase. Tester^D decompiles the sketch skt into $x_1, x_2, \ldots, x_m$, then it runs Equal^{D,D^*} locally. Each time Equal calls the oracle $\mathcal{O}_{D^*}$, Tester uses the next x_i as the result. Tester outputs the output of Equal.

Correctness and soundness follow directly from the properties of the Equal algorithm. The sketch is of size $\tilde{O}(n^{2/3})$, and the tester uses $\tilde{O}(n^{2/3})$ samples from D, satisfying succinctness.

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